

# Influence of Temperature on Fatigue Life of Lock Test Mechanism

Yaojia Han

*School of Aeronautics  
Northwestern Polytechnical  
University  
Xi'an, China  
hyjnwpu@163.com*

Yuanyuan Zhao

*School of Aeronautics  
Northwestern Polytechnical  
University  
Xi'an, China  
2570019015@qq.com*

Dingnan He

*School of Aeronautics  
Northwestern Polytechnical  
University  
Xi'an, China  
hedingnan@mail.nwpu.edu.cn*

Xiangyang Li

*School of Aeronautics  
Northwestern Polytechnical  
University  
Xi'an, China  
asdfly331@mail.nwpu.edu.cn*

Deyin Jiang

*School of Aeronautics  
Northwestern Polytechnical  
University  
Xi'an, China  
jiangdy@mail.nwpu.edu.cn*

Weimin Cui

*School of Aeronautics  
Northwestern Polytechnical  
University  
Xi'an, China  
cuiwm@nwpu.edu.cn*

**Abstract**—Aircraft cabin door lock is very important in flight missions. The reliability test of a lock mechanism is carried out. Firstly, a test mechanism that meets the test requirements is designed, the fatigue life analysis method of key parts of the mechanism is determined, and the fatigue life analysis framework of the mechanism is proposed. Then LMS.Virtual.Lab software is used to analyze the multi-body dynamics of the test mechanism, and the load spectrum of the analyzed object is obtained. Based on the dynamic analysis and the five-block diagram of fatigue simulation analysis, the fatigue life of the analyzed object was analyzed by Ncode DesignLife software. Finally, the thermal and structural coupling analysis of the test mechanism was carried out by using ANSYS software. After obtaining the stress data, the fatigue life of the analyzed object under the coupling of temperature and mechanical load was studied, and the influence of temperature on the fatigue life of the lock test mechanism was analyzed.

**Keywords**—test mechanism; kinetic analysis; fatigue life; thermal structure coupling

## I. INTRODUCTION

Lock mechanisms are widely used in aircraft mechanical products. For the reliability test of the lock mechanism, it is necessary to design the test organization, which requires the test organization to meet the environmental requirements of the test product while fixing the test product and transmitting the load. Therefore, it is necessary to analyze the durability of the designed test organization in consideration of environment and loading. For the test organization in this article, it is mainly subject to fatigue durability, which is caused by cyclic loading. At present, there are two main methods to study the fatigue life of structures at home and abroad. One is to adopt the physical failure method for design, and the other is to directly predict the fatigue life, that is, to predict the stress load spectrum of the

structure and the S-N curve of the material by equivalent comparison to determine its life. Compared with the fatigue design method under static strength and conventional fatigue design method, the fatigue life study under dynamic load can reflect the actual working environment of the structure.

The stress-life method is usually used in engineering to predict fatigue life, and it is widely used in anti-fatigue design. N.Hamsaei and A.Fatemi [1] developed a set of fatigue life prediction software, which can define various parameters of materials, such as strength coefficient, strain hardening index and other material parameters, to calculate the fatigue life under multi-axis. The fatigue life predicted by the program is close to the experimental results. J.Kramberger [2] and S.Glodez [3] calculated the gear-tooth root bending fatigue life based on finite element technology and strain-life method, and simulated the tooth root crack propagation using a simple Paris formula. Wonjong Kang [4] comprehensively considered the impact of stamping and welding on the components, based on the finite element analysis method, carried out a fatigue analysis on a certain type of automobile suspension components, and the results showed that compared with stamping, welding has a greater impact on the fatigue life of components. Literature by above knowable, the current domestic and external research on fatigue life prediction of mechanical system, the multi-body dynamics method with the finite element method can predict the fatigue life of high, has been widely applied, and then verify the validity of the method by way of test, in view of the fatigue life prediction problem institutions key-module, scholars have conducted research is still relatively small. This article focuses on the fatigue problem of the mechanism.

Due to the structural thermal flutter and instability caused by thermal-structural coupling problems, such as structural failure and other problems, thermal-structural coupling has been

studied more and more, especially in the field of aerospace and other important significance and influence. Boely [5-6] studied the thermal response analysis of a cantilever beam under a sudden load of heat flow, and found that heat may induce vibration, which laid the foundation for future research on thermal flutter and thermal vibration. Thornton E A [7] has been studied by many scholars on thermal flutter under thermal-structure coupling. The research objects are large aircraft and space structures, and this study has become the main theoretical basis for the research on thermal-induced vibration and thermal flutter.

## II. DESIGN OF TEST MECHANISM

According to the reliability analysis process of the test mechanism, the initial design of the test mechanism is considered from the following points. The cabin door lock is fixed, the cabin door lock needs to be installed above the lock ring, which can be triggered under the movement of the lock ring. Simulation of cabin door movement, the design of the test mechanism needs a mechanism that can simulate the cabin door movement, and the key point is that it can trigger the lock hook movement. Load drive of the test mechanism, the design of the load drive of the test mechanism should be able to achieve variable load, the driving function needs to use hydraulic actuating cylinder, and the load of the force uses spring loading to meet the requirements of variable load and locking load. Force transmission of the test mechanism, the force transmission mechanism plays an important role in the whole test. It is necessary to connect and load the driving module and the door motion simulation module to accurately transfer the force to the lock ring in the door motion simulation module, so that the force of the lock ring on the lock hook can meet the design requirements of the test scheme. The basic form of the preliminary design test mechanism is shown in Figure 1.

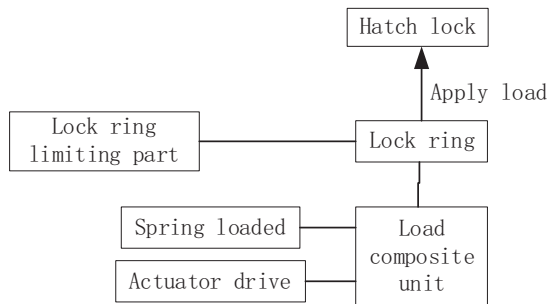


Fig. 1. The basic function block diagram of the test mechanism.

According to the above functional module division of the test object, the design difficulty of the test mechanism is the load compound component module, which needs to transfer the force to the cabin door lock hook, and the module needs to meet the requirements of the load and loading at different working stages and has a rotational degree of freedom. Specific design instructions for different modules are as follows:

### A. Fixed Module

The fixed module mainly includes the bottom plate, the hatch door lock mounting plate, the strengthening plate and the

mounting support of other modules. The hatch lock mounting plate is vertically consolidated on the upper side of the bottom plate, and the upper part is provided with a bolt hole for installing the hatch lock, through which the test object is installed on the hatch lock mounting plate. The stiffening plate is consolidated with the hatch door lock mounting plate and bottom plate, as shown in Figure 2.

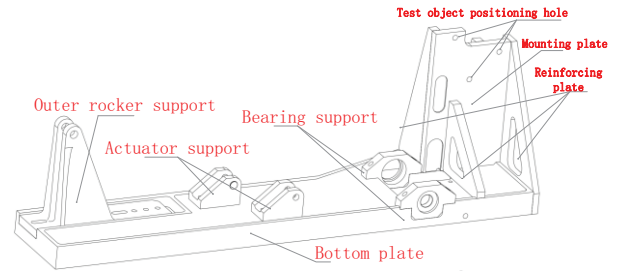


Fig. 2. The basic function block diagram of the test mechanism

### B. Load the Driver Module

The loading drive module is mainly composed of an actuating cylinder, a guide block, a No. 1 slide bar, a No. 2 slide bar, a No. 1 spring, a No. 2 spring, a spring limit plate, and a load adjusting nut, as shown in Figure 3. In order to make the loading force meet the requirements of the test scheme, the spring parameters are determined as follows.

The maximum load provided by the latching loading spring is 2400N and the compression deformation is 15mm. The spring coefficient is required to be  $2400/15=160\text{N/mm}$ . The blue light load die spring with inner diameter of 30mm and outer diameter of 60mm is selected. The ultimate pressure is 4763N and the compression is 40%. The required spring length is  $4763\text{N}/(160\text{N/mm})/0.4=74.4\text{mm}$ , and the free length is rounded to 75mm.

The maximum load of the reverse spring is  $5933-2400=3533\text{N}$ . In order to reduce the spring elasticity fluctuation during the test, the length of the spring should be as long as possible. The amount of compression in the locking process should be close to  $50-15=35\text{mm}$ . Therefore, the elastic coefficient is  $3533/35=100\text{N/mm}$ . The blue light load die spring with inner diameter of 30mm and outer diameter of 60mm is selected. The spring free length required is  $4763\text{N}/(100\text{N/mm})/0.4=119.1\text{mm}$ . To facilitate the installation of the test fixture (there is no need to apply pre-pressure on the spring), the free length is rounded to 100mm.

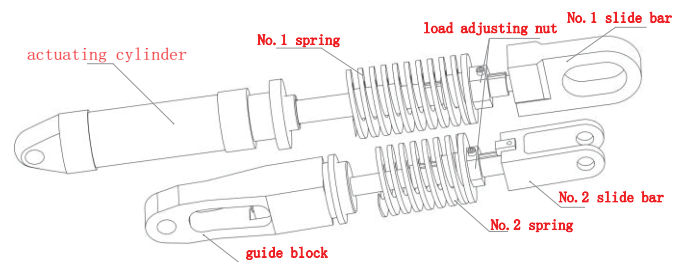


Fig. 3. Stereo diagram of loading driver module.

### C. Load Transfer Module

The load transfer module is mainly composed of L-shaped rocker arm, adjustable rocker arm, rotating shaft, bearing, connecting rod, etc. The rotating shaft is installed in the bearing seats at both ends, and the adjustable rocking and L-shaped rocker arm are fixed on the rotating shaft. One end of the connecting rod is connected with one end of the L-shaped rocker arm, and the other end is connected with the hatch door simulation module, as shown in Figure 4.

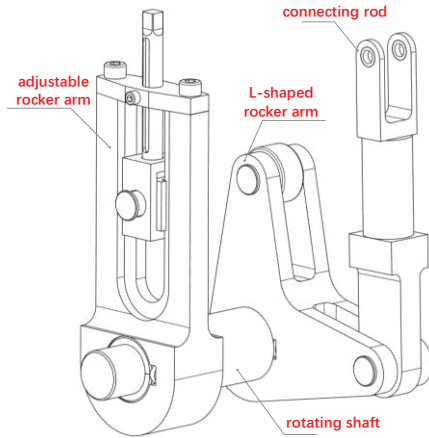


Fig. 4. Stereo diagram of load transfer module.

### D. Door Simulation Module

The hatch door simulation module is mainly composed of locking ring, inner rocker arm, outer rocker arm, fixing bolt, positioning pin, etc. One end of the inner rocker arm is provided with a pair of ear pieces, two ear pieces are arranged in the middle of a locking ring through a rotating pair, the other end of the outer rocker arm is provided with a chute, the bottom of the chute is provided with a plurality of through holes; One end of the outer rocker arm is installed in the chute of the inner rocker arm, and the other end is connected with the rocker arm support through a rotating pair. The radius of swinging up and down of the lock ring is adjusted and controlled through the positioning hole. After the No. 1 slide rod is retracted in place, the lock ring just triggers the hatch lock, as shown in Figure 5.

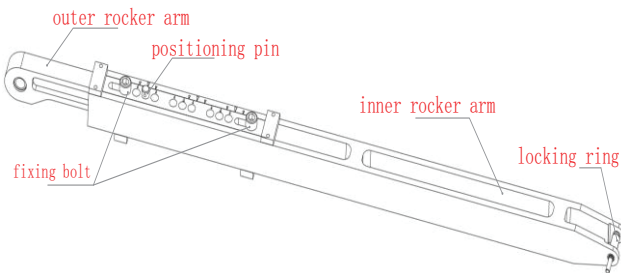


Fig. 5. Schematic diagram of hatch simulation module.

The overall structure of the test mechanism is shown in Figure 6.

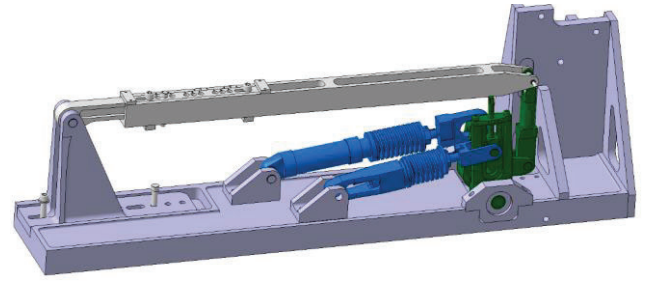


Fig. 6. Structure diagram of test mechanism.

## III. FATIGUE DURABILITY ANALYSIS METHOD OF TEST MECHANISM

### A. Fatigue Cumulative Damage Theory

Fatigue life refers to the number or time of action of cyclic load from the beginning of work to the destruction of material or structure under the action of alternating load. According to the different load types, the fatigue types can be generally divided into the following four types: mechanical fatigue, thermal fatigue, corrosion fatigue, rolling contact fatigue and sliding contact fatigue. The fatigue cumulative damage theory is the basis of fatigue life analysis. For fatigue caused by constant amplitude load, the fatigue life of the structure under different stress levels can be directly calculated by S-N curve of the material. Any fatigue cumulative damage theory must be based on the definition of fatigue damage  $D$  and on the evolution of fatigue damage. The Miner linear cumulative damage theory is used to predict the fatigue life in this paper. It is simple and convenient, and can predict the fatigue life well.

Damage caused by one cycle:

$$D = \frac{1}{N} \quad (1)$$

Damage caused by  $n$  cycles under constant amplitude load:

$$D = \frac{n}{N} \quad (2)$$

Damage caused by  $n$  cycles under variable amplitude load:

$$D = \sum_{i=1}^n \frac{1}{N_i} \quad (3)$$

Where  $N_i$  is the fatigue life of the current load level  $S_i$ .

Critical fatigue damage  $D_{CR}$ : If it is a constant amplitude cyclic load, when the number of cyclic loads  $n$  is equal to the fatigue life  $N$ , the material or structure will undergo fatigue failure, when  $n = N$ ,  $D = D_{CR}$ , that is:

$$D_{CR} = 1 \quad (4)$$

### B. Description of Fatigue

In general, high cycle fatigue is caused by elastic strain, and the fatigue life is generally above  $10^6$ , which is relatively long. The fatigue caused by plastic strain plays a major role is strain fatigue, and the fatigue life is generally less than  $10^4 \sim 10^5$ , and the life is relatively short.  $S - N$  curve is used to reflect the

relationship between external stress and fatigue life, and is obtained by fitting the fatigue test results under many different stress ranges. Because this kind of curve was first proposed by German Wohler, it is also called Wohler curve. A typical  $S - N$  curve is shown in Figure 7. The abscissa is the number of stress cycles, and the ordinate is the stress level. According to the number of stress cycles, the curve can be divided into three sections: low cycle fatigue zone (LCF), high cycle fatigue zone (HCF) and sub-fatigue zone (SF).

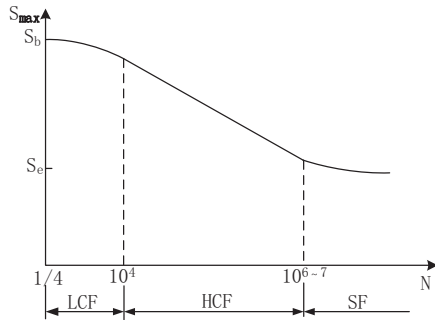


Fig. 7. Typical S-N curve.

### C. Determination Method of Fatigue Load Spectrum

The load spectrum is an important factor to predict the fatigue life of the test mechanism. In this paper, the fatigue load spectrum is determined by calculating the load history of the fatigue key parts through the multi-body dynamics model, and then processing the load history curve according to the rain-flow counting method to get the load spectrum of the fatigue key parts. The rainflow counting method was proposed by the engineers Matsuiski and Endo in 1968. It belongs to the two-parameter cycle counting method and is the most widely used method for fatigue design and failure assessment at present. Figure 8 below shows the load processing mode of rainflow counting method, which can be referred to [8] for details.

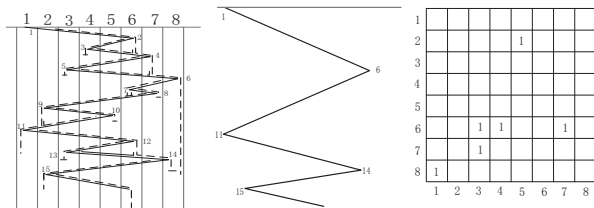


Fig. 8. Schematic diagram of rain flow counting method.

## IV. MULTI-BODY DYNAMICS ANALYSIS OF THE TEST MECHANISM

### A. Relationship Between the Components of The Mechanism

In this paper, the LMS Virtual. Lab Motion module is used to simulate the multi-body dynamics of the test Mechanism. The designed components are imported into the Mechanism Design platform under the software as rigid bodies to determine the relative Motion between rigid bodies and define the constraints,

spring attributes, forces and initial conditions between components. The kinematic relationship between the components is defined by the kinematic pair (Joint), which is shown in Table 1. In this paper, the sphere-tensile surface is used to define the kinematic relationship between the locking hook and the locking ring and the kinematic relationship between the L-shaped rocker arm in the U-shaped hole of the No.1 slider, as shown in Figure 9 and Figure 10.

TABLE I. DYNAMICS SIMULATION OF EACH MOTION PAIR

| Body1               | Body2             | Motion pair      |
|---------------------|-------------------|------------------|
| Actuator support    | Actuator          | Rotating pair    |
| Guide block support | Guide block       | Rotating pair    |
| Actuator            | No. 1 slider      | Mobile pair      |
| Guide block         | No. 2 slider      | Cylindrical pair |
| Bearing seat        | Bearing           | Rotating pair    |
| No. 2 slider        | Adjustable rocker | Rotating pair    |
| L-shaped rocker     | Link              | Rotating pair    |
| link                | Lock ring         | Rotating pair    |
| Inside rocker       | Rocker support    | Rotating pair    |
| Lock ring           | Mounting holes    | Rotating pair    |
| Shackle             | Mounting plate    | Rotating pair    |

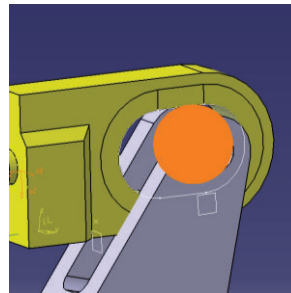


Fig. 9. No. 1 slide bar contacts the L - shaped rocker arm ball tensile surface.

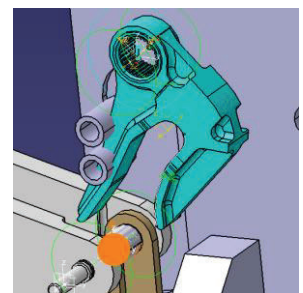


Fig. 10. Stretch contact between locking ring and locking hook ball.

Applying drive: according to the motion condition of the test mechanism in actual work, from the locking to triggering stage, a mobile drive is applied on the actuator cylinder and the moving pair of No. 1 slide rod to simulate the hydraulic system to exert the locking fluid pressure on the tested product, so that the locking ring triggers the locking hook movement to the triggering Angle stage; From the triggering stage to the locking stage, a drive is applied on the rotating pair of the locking hook and the mounting plate to simulate the action of liquid pressure on the locking hook, so that the locking ring can be locked under the action of the locking hook. In this paper, the inverse dynamics method is used to analyze the force of the test mechanism, that is, to find the internal force between the components under the condition of known movement of the mechanism.

## B. Simulation of Multi-rigid Body Dynamics System

The multi-rigid body dynamics analysis and simulation of the test mechanism verify the rationality of the design mechanism and the established simulation model, which is the premise of the multi-flexible body system simulation and fatigue simulation calculation. The movement cycle of the test mechanism is 3s. In this paper, a complete simulation cycle is set, the simulation step size is set as 0.01s, and the analysis type is set as DYNAMIC.

After the creation of the previous constraints, the imposition of load and drive, and the setting of solving parameters are completed, the solution is carried out. LMS software will solve the multi-body dynamics equation based on the established constraint relationship and load of the simulation model and the multi-body dynamics theory method of the multi-rigid body system. As shown in Figure 11 to 13, they are respectively the kinematic load spectrum at the connection between L-shaped rocker arm and No.1 slider bar, the kinematic load spectrum at the connection between adjustable rocker arm and No.2 slider bar, and the kinematic load spectrum at the connection between L-shaped rocker arm and connecting rod.

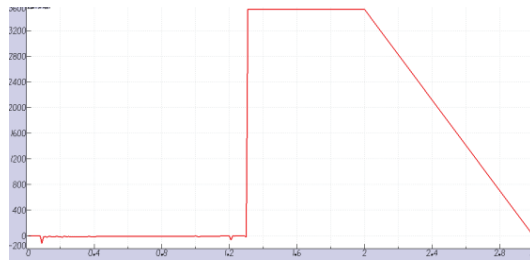


Fig. 11. Load spectrum of the kinematic pair at the connection between the L-shaped rocker arm and the No. 1 slide bar.

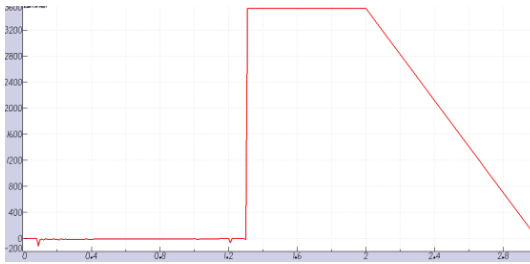


Fig. 12. Load spectrum of the kinematic pair at the connection between the adjustable rocker arm and the No. 2 slide bar.

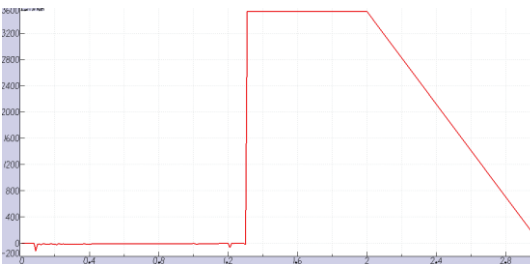


Fig. 13. Load spectrum of kinematic pair at the connection between L-shaped rocker arm and connecting rod.

In this multi-rigid body simulation, the condition of the trigger stage was not considered, because the force in the trigger stage was small, and the fatigue damage caused could be ignored. Therefore, it can be seen in the load spectrum that the 0-1s force is very small. However, in the motion load spectrum of the joint between the L-shaped rocker arm and the No.1 slide bar, the maximum value is 142N, which is caused by the contact motion between the U-shaped hole and the No. 1 slide bar. After 1S, the lock hook starts to move. After 1.18s, it contacts the lock ring and drives the lock ring to move together to complete the locking process.

## V. FATIGUE LIFE PREDICTION OF TEST MECHANISM

In this paper, the fatigue life estimation method based on the finite element is adopted for the fatigue life analysis of the fatigue key components. The method of the five-block diagram of fatigue analysis is shown in Figure 14. Firstly, the finite element analysis is carried out. According to the actual working conditions of the flexible body, the transient analysis is adopted. The force is applied by the load spectrum corresponding to the motion pairs extracted from the upper section. Then the force is applied to the corresponding position to ensure the authenticity of the constraint and force application simulation.

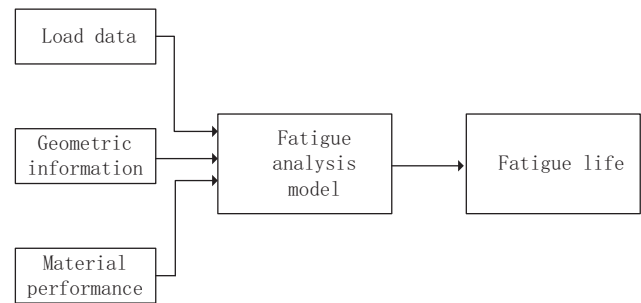


Fig. 14. Five block diagram of fatigue analysis.

Due to the test condition is more complex, in different stages are different form the size of the load, but after strength analysis on stiffness of the analysis object, specimen in preparation for the closure, trigger phase force is small, in line with the infinite life thoughts, less damage, so can not included in this stage when fatigue calculation applied load. In the fatigue life analysis, the load spectrum is applied mainly in the locking stage, the locking holding stage and the unlocking stage.

In this paper, NCODE software and ANSYS software are used for co-simulation, and the results of finite element analysis are directly imported into the fatigue life analysis software to calculate the fatigue life. In Ncode, the drag and drop modeling method is adopted to build the fatigue analysis block diagram, as shown in Figure 15, which includes the input module of the finite element model, the material endow module, the fatigue analysis module, the hot spot detection function, the cloud image display function, and the node data information after the fatigue analysis calculation is completed.

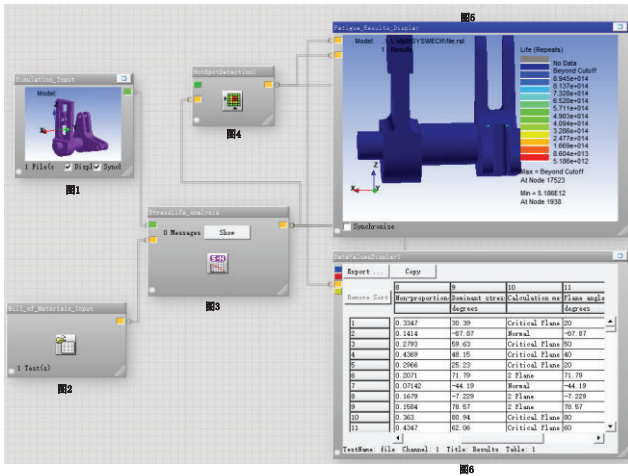


Fig. 15. Fatigue analysis module.

#### A. Basic Settings for Fatigue Analysis

The solution setting of fatigue analysis is mainly carried out in 3 Advanced Edit, including the setting of surface parameters of parts, stress combination mode, stress correction, survival rate and so on. In this paper, the VonMises equivalent stress combination with sign is adopted, and Goodman's rule is used for stress correction.

#### B. Material Information Setting

Material setting includes material selection or creation, material parameter input and so on. The material used in this paper is structural steel. Based on the basic parameters of structural steel, Generate is used to create the S-N curve of the generated material, and the obtained S-N curve is shown in Figure 16.

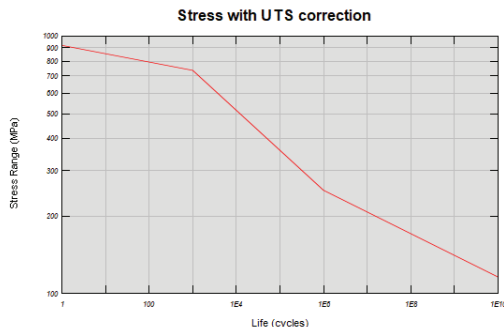


Fig. 16. S-N curve of material.

#### C. Load Information Setting

The finite element analysis in this paper is a transient dynamic analysis, and the stress obtained is the stress related to time. Therefore, the load directly adopts the load from the finite element. The load setting is shown in Figure 17.

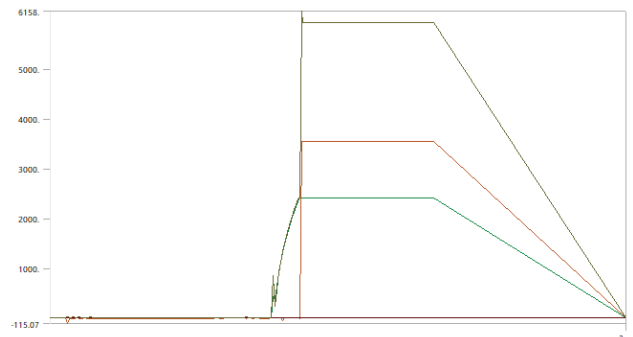


Fig. 17. Fatigue solution load spectrum.

#### D. The Results Show

In fatigue calculation, the damage and life of each time step are calculated step by step, and then linear superposition is carried out. After the calculation is completed, there are two display results: one is damage and life display cloud map, as shown in Figure 18 to 20; the other is data list, which can clearly see the fatigue damage or fatigue cycle times under each node number.

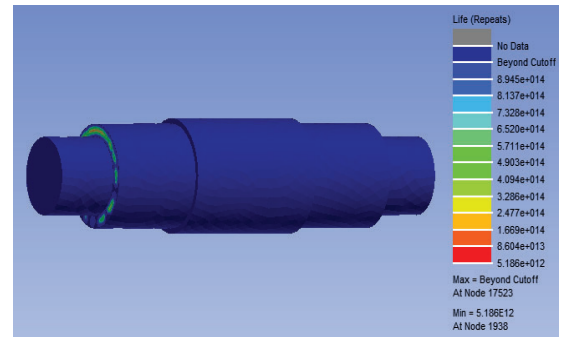


Fig. 18. Shaft life cloud diagram.

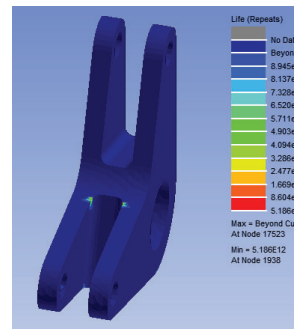


Fig. 19. L-shaped rocker arm life cloud map.

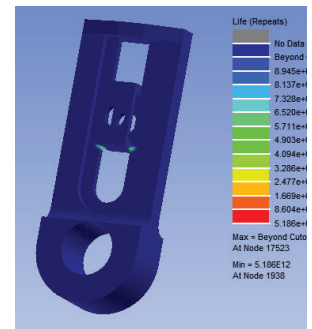


Fig. 20. Cloud diagram of adjustable rocker arm life.

As can be seen from the figure, the overall life of the analyzed object is large, which meets the test requirements. The part with the minimum fatigue life appears on the sliding block of the adjustable rocker arm. The life is smaller than that of other parts due to the stress concentration on the sliding block.

## VI. FATIGUE LIFE SIMULATION UNDER TEMPERATURE EFFECT

### A. Thermal-structural Coupled Finite Element Model

Because the test object must consider the temperature factor in the test, it is necessary to consider the influence of temperature on the durability of the test mechanism in the design of the test mechanism. Considering that the small deformation of the structure will not affect the results of thermal analysis, the sequential coupling method is used for the simulation of the fatigue key components. Firstly, the temperature field of the key component is simulated, and then the temperature field is imported into the transient dynamics calculation as the volume load for dynamic simulation analysis. The sequential coupling process built by ANSYS is shown in Figure 21.

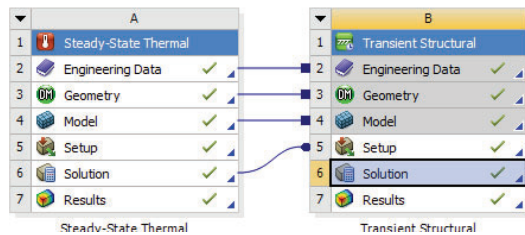


Fig. 21. Thermal-mechanical coupling flow chart of sequential coupling.

The thermal deformation of the analysis object under the influence of temperature will produce thermal stress in the constrained position. However, in the working process of the analysis object, the adjustable rocker arm and L-shaped rocker arm are installed on the rotating shaft and the two ends of the rotating shaft are respectively constrained by the rotating shaft support, thus generating thermal stress. The linear strain is generated under the influence of temperature, and the temperature load of the structure is obtained by taking the linear strain as the initial strain, and then the stress under the thermal-structural coupling is obtained by combining the temperature load with other loads according to the elastic mechanics equation.

This paper selects the lowest temperature and the highest temperature of the temperature loading test to carry out finite element analysis on the analysis object, and the finite element results are shown in Figure 22 to 25.

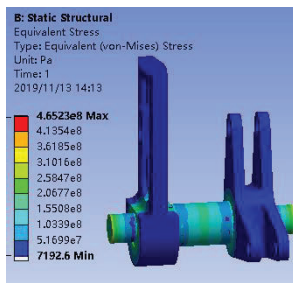


Fig. 22. Stress distribution of the analyzed object at low temperature.

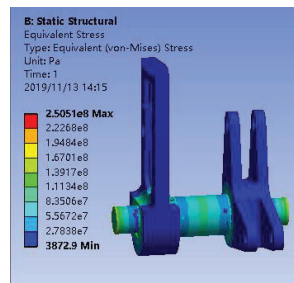


Fig. 23. Stress distribution of the analysis object at high temperature.

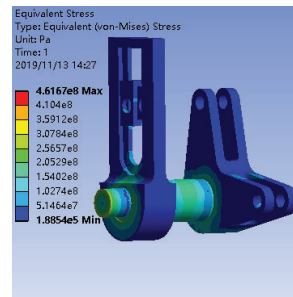


Fig. 24. Thermal-structural coupling stress distribution at low temperature.

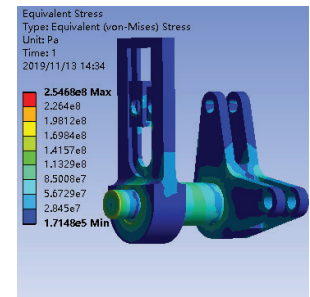


Fig. 25. Thermal-structural coupling stress distribution at high temperature.

For fatigue life calculation with temperature in mind, the calculation process is the same as in the previous section. The difference is that the temperature field that needs to be imported into the analysis object is imported into DesignLife together with the stress calculation results, and the analysis module is established, as shown in Figure 26. The calculated life cloud diagram of the analyzed object is shown in Figure 27 to 32.

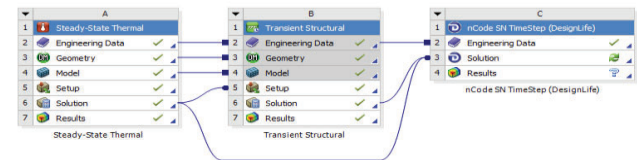


Fig. 26. Flow chart of thermal-structural coupling fatigue life calculation.

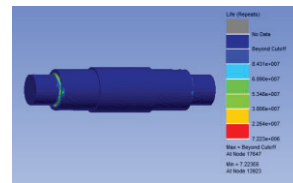


Fig. 27. Fatigue life cloud of rotating shaft at low temperature.

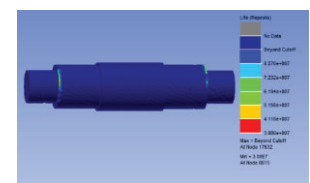


Fig. 28. Fatigue life cloud of rotating shaft at high temperature.

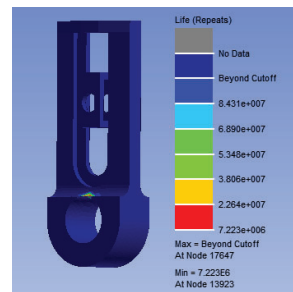


Fig. 29. Cloud diagram of adjustable rocker arm life at low temperature.

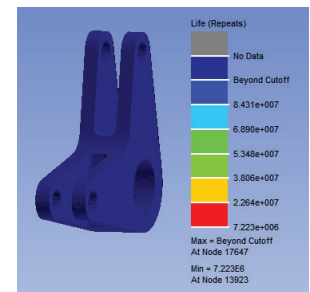


Fig. 30. Cloud diagram of L-shaped rocker arm life at low temperature.

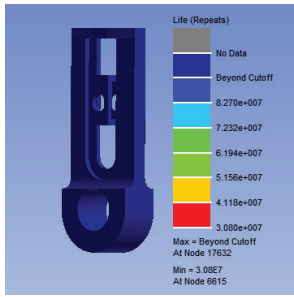


Fig. 31. Cloud image of adjustable rocker arm life at high temperature.

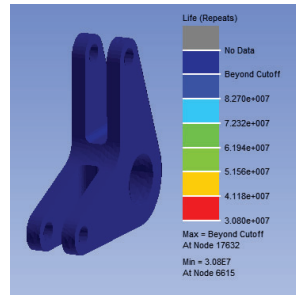


Fig. 32. Cloud image of L-shaped rocker arm life at high temperature.

Due to the great influence of temperature on the ultimate tensile strength of materials, the S-N curve of materials used is different from that of actual materials. Therefore, several groups of different ultimate tensile and compression strengths are selected for calculation in this paper. The relationship between fatigue life and ultimate tensile and compression is shown in Figure 33 and Figure 34.

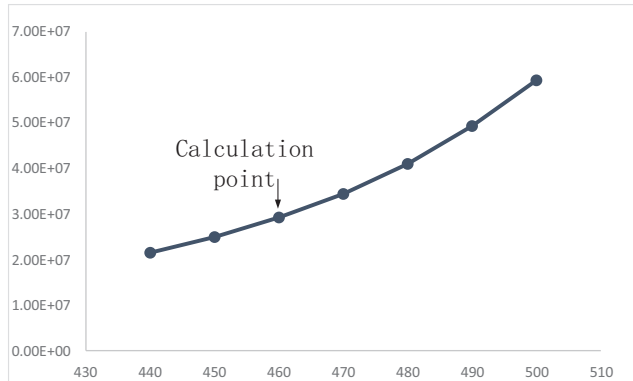


Fig. 33. Fatigue life of different tensile limits at low temperature.

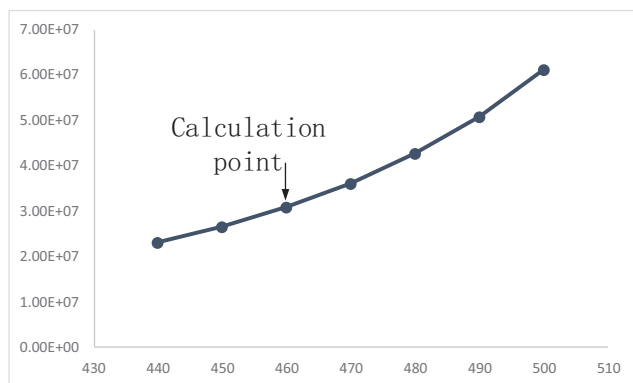


Fig. 34. Fatigue life of different tensile limits at high temperature.

## VII. CONCLUSION

According to the actual test requirements, this paper has completed the design of the test mechanism for the reliability test of the landing gear cabin door lock of a certain type of aircraft. The framework of fatigue life analysis, the method of fatigue life prediction and the load spectrum of mechanism are determined. The multi-body dynamics simulation of the test mechanism was carried out, and the load spectrum of the object to be analyzed was obtained, which was used for subsequent loading of fatigue simulation. Combined with the five-block diagram of fatigue simulation analysis, the fatigue life of the analyzed object was predicted by the fatigue calculation software DesignLife. It can be seen from the calculation results that at normal temperature, the fatigue hot spots of the test mechanism concentrate on the lower corner of the slider of the adjustable rocker arm. Using ANSYS software, the thermal and structural coupling analysis of the test mechanism was carried out. After obtaining the stress data, the life of the analyzed object under the coupling of temperature and mechanical load was discussed. The results show that the temperature has a great influence on the fatigue life of the analyzed object under the influence of thermal structure coupling. When the analyzed object is in high or low temperature, the adjustable rocker arm and the rotating shaft, and the contact part of L-shaped rocker arm and the rotating shaft has a large force, which reduces the fatigue life. Because the temperature change has a great influence on the ultimate tensile strength of materials, the fatigue life under different tensile limits is calculated in this paper. It is found that the ultimate tensile strength of materials is an important factor affecting the fatigue life.

## REFERENCES

- [1] N. Shamsaei and A. Fatemi "Fatigue Life Predictions under General Multiaxial Loading Based on Simple Material Properties," *SAE International Journal of Materials and Manufacturing*, vol. 4, no. 1, pp. 651-658, 2011.
- [2] J. Kramberger, M. Sraml, and I. Potrc, "Numerical calculation of bending fatigue life of thin-rim spur gears," *Engineering Fracture Mechanics*, vol. 71, pp. 647-656, 2004.
- [3] S. Glodez, M. Sraml, and J. Kramberger, "A computational model for determination of service life of gear," *International Journal of Fatigue*, vol. 24, no. 10, pp. 1013-1020, 2002.
- [4] Woojong Kang, Gihoon Kim, Keejune Moon, and Taejun Kim. "Manufacturing Effects on the Fatigue life Assessment by Finite Element Methods," *SAE Technical Paper*, 2007, doi: <https://doi.org/10.4271/2007-01-3696>.
- [5] B. A. Boley, *Theory of thermal stresses*, NewYork: John Wiley & Sons, Inc., 1961.
- [6] B. A. Boley, "Thermal induced vibration of beams," *Journal of the Aeronautical Sciences*, vol. 23, no. 2, pp. 179-181, 1956
- [7] E. A. Thornton, *Thermal structures for aerospace applications*, NewYork: AIAA, 1996.
- [8] H. Li, "Calculation of Fatigue Damage of Nuclear Power Plant Component Material under Random Load by Rain Flow Method," M.S. thesis, China Institute of Atomic Energy, Beijing, 2001.